

Simultaneous exposure of non-diabetics to high levels of dioxins and mercury increases their risk of insulin resistance.



Insulin resistance and the defective function of pancreatic β -cells can occur several years before the development of type 2 diabetes. It is necessary to investigate and clarify the integrated effects of moderate-to-high exposure to dioxins and mercury on the pancreatic endocrine function. This cross-sectional study investigated 1449 non-diabetic residents near a deserted pentachlorophenol and chloralkali factory. Metabolic syndrome related factors were measured to examine associations with serum dioxin and blood mercury. We also investigated associations between insulin resistance (HOMA-IR > 75th percentile), defective pancreatic β -cells function (HOMA β -cell > 75th percentile), serum dioxins and blood mercury. After adjusting for confounding factors, we found that insulin resistance increased with serum dioxins ($b = 0.13$, $P < 0.001$) and blood mercury ($b = 0.01$, $P < 0.001$). Moreover, participants with higher serum dioxins or blood mercury were at a significantly increasing risk for insulin resistance ($P(\text{trend}) < 0.001$). The joint highest tertile of serum dioxins and blood mercury was associated with elevated HOMA-IR at 11 times the odds of the joint lowest tertile (AOR 11.00, 95% CI: 4.87, 26.63). We hypothesize that simultaneous exposure to dioxins and mercury heightens the risk of insulin resistance more than does individual exposure. Copyright © 2010 Elsevier B.V. All rights reserved.